Technology Stack

Date	10 July 2026
Team ID	xxxxxx
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	2 Marks

Define Your Technology Stack:
Define the technologies used to design, develop, train, and deploy the Rising Waters: A Machine Learning Approach to Flood Prediction system. The selected technology stack supports machine learning model development, data analysis, web application development, and deployment, ensuring an accurate, scalable, and user-friendly flood prediction solution.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side** *(e.g., React, Angular, HTML/CSS)*	*HTML5, CSS3, JavaScript, Bootstrap*	*Used to build a responsive and interactive web interface for entering weather parameters and displaying flood prediction results. (templates/, static/css, static/js)*

2	**Backend / Server-Side** *(e.g., Node.js, Python/Django, Java)*	*Python, Flask*	*Flask handles routing, user requests, prediction logic, and communication between the frontend and the trained machine learning model. (app.py)*
3	**Database / Data Storage** *(e.g., MySQL, MongoDB, PostgreSQL)*	*Excel (.xlsx), Pickle (.pkl), JSON*	*Historical rainfall datasets are stored in Excel files, while trained models, scaler, feature columns, and feature statistics are stored as Pickle and JSON files. (dataset/, models/)*
4	**Cloud / Hosting / Deployment** *(e.g., AWS, Azure, Heroku, Docker)*	*Heroku / Render (using Procfile & runtime.txt)*	*Enables deployment of the Flask application, making the flood prediction system accessible through a web browser. (Procfile, runtime.txt)*
5	**Version Control & CI/CD**	*Git, GitHub*	*Used for source code management, version tracking, collaboration, and maintaining project history.*

	(e.g., GitHub, GitLab, Jenkins)		
6	**Third-Party APIs / Other Tools** *(e.g., Stripe, Twilio, Google Maps)*	*Scikit-learn, Pandas, NumPy, Matplotlib, Joblib, Jupyter Notebook*	*Used for data preprocessing, exploratory data analysis (EDA), machine learning model training, visualization, model serialization, and experimentation. (notebooks/eda_and_modeling.ipynb)*